

Ethan Moon

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Computer science engineering student and full-stack engineer building production software, scientific computing systems, and human-centered products. Experience spans TypeScript product platforms, React/Expo interfaces, PostgreSQL APIs, computer vision, and computational fluid dynamics.

EDUCATION

B.S.E. Computer Science, College of Engineering — University of Michigan | Ann Arbor, MI

Expected May 2028

Coursework: Computers & Programming, Discrete Math, Proof-Based Linear Algebra, Multivariable Calculus, Differential Equations, Statistics, Physics

TECHNICAL EXPERTISE

Software: TypeScript, JavaScript, Python, C++, SQL; React, Next.js, React Native/Expo, Node.js, Hono

Data & delivery: PostgreSQL, Drizzle ORM, OpenAPI/Orval, Docker, Git, GitHub Actions, REST APIs

Engineering & research: OpenFOAM/RheoTool, ParaView, NASA GMAT, Octave/MATLAB, Arduino/ESP32, computer vision, digital twins, experimental design

Languages: English and Korean (native/bilingual); Spanish (limited working)

ENGINEERING EXPERIENCE

Lead Engineer & Head of U.S. Team — Ody

May 2026 – Present

- Shipped 107 merged PRs in 10 weeks across six product apps and the backend, delivering database schemas, APIs, generated clients, web interfaces, and mobile workflows.
- Built four merchant-facing feature areas, including AI-assisted multilingual menus, email campaigns, and multi-location analytics, with production-grade authorization and validation.
- Worked across PostgreSQL/Drizzle, Hono, OpenAPI/Orval, React, and Expo in a shared TypeScript monorepo.

Co-Author Researcher — Grand Valley State University

Sep 2025 – Present

- Developed and calibrated non-Newtonian, multiphase CFD models for 32 bioink-deposition conditions in OpenFOAM/RheoTool, achieving 96% agreement with experimental data.
- Built a Dockerized Python/JavaScript analysis platform with VTK-to-CSV pipelines and interactive visualizations for 15,300+ records across 51 timesteps per simulation.
- Programmed real-time nozzle-anomaly detection using computer vision and digital-twin modeling; co-authored a peer-reviewed ASEE conference paper and is extending the work for journal submission.

Summer Researcher — Grand Valley State University

Jun 2025 – Aug 2026

- Orbital dynamics: simulated transfers, rendezvous, perturbations, and fuel-efficient satellite-formation maneuvers in Octave and NASA GMAT.
- Aerodynamic hysteresis: designed and 3D-printed airfoil prototypes, instrumented wind-tunnel tests with Arduino sensors, and compared results with CFD simulations.

Simulation & Robotics Engineering Intern — Casual Simulation

Feb 2024 – Jan 2026

- Developed a VR rowing-performance simulation adopted by a local high-school rowing team and iterated on Arduino-based robotic-dog controls, improving stability by 21%.

Advanced Math Instructor — Mathnasium

Dec 2025 – Present

- Teach algebra, geometry, and precalculus through individualized explanations and targeted practice plans; mentor students in mathematical reasoning and problem-solving confidence.

SELECTED ENGINEERING PROJECTS

No Spoilers — Full-Stack Web App & Chrome Extension — TypeScript, React, PostgreSQL

2026 – Present

- Built spoiler-free movie discovery and watchlist workflows spanning 120 API handlers, 43 PostgreSQL models, and 42 user-facing page entry points.
- Implemented a privacy-first browser extension with on-device spoiler classification and watchlist sync; the repository currently passes 54 automated tests.

ViolinTwin — AI Practice Coach — React Native/Expo, Swift, Kotlin

2026

- Prototyped a mobile practice coach with native audio-analysis scaffolding, recording workflows, pitch/rhythm/note-timing estimation, and MusicXML/MIDI targets.
- Designed an offline-capable fallback analysis path and reproducible audio-analysis tests.

Smart Plate — Accessible Nutrition App — Defold, Lua, BLE-ready architecture

2023 – 2025

- Designed low-stimulation, gamified food-exposure workflows with rewards, recommendations, progress tracking, and parent-facing feedback.
- Prototyped load-cell and microcontroller integration; a real-world pilot increased willingness to try new foods.

Helping Hands Soccer Training Platform — Web/mobile product

2022 – 2026

- Built location- and goal-based coach/player matching plus progress dashboards for a volunteer nonprofit that donated \$2,400 in scholarships.

RESEARCH & PUBLICATIONS

Closed-loop bioprinting and digital-twin research. Co-authored and presented peer-reviewed work at the ASEE North Central Conference; continuing the CFD, vision, and controls work for a bioprinting journal manuscript.

Alternative aviation fuels. Authored comparative research on sustainable replacements for aviation kerosene, published through the International STEM Research Journal.

LEADERSHIP & COMMUNITY

Founder & Head Coach — Helping Hands, N.P.O.

Feb 2022 – Apr 2026

- Founded a volunteer soccer-training nonprofit, recruited 10+ high-level coaches across the United States, and built the program's matching platform and operating workflows.
- Directed coaching and fundraising that provided \$2,400 in scholarships to young athletes.

Program & Exhibit Director — Grand Rapids Public Museum

May 2025

- Proposed and sourced a 25+ image astrophotography exhibit that the museum adopted as an interactive discovery cart; presented astronomy content to visitors.

Captain & Starting Defender — Forest Hills Central Varsity Soccer

2024 – 2026

- Led team strategy and culture while earning All-Conference, All-District, All-Region First Team, and All-State Honorable Mention recognition.

SELECTED HONORS

- DECA State Champion, Integrated Marketing Campaign — Service (2026)
- Second Place Overall and Grand Prize Honorable Mention, Southwestern Michigan Science & Engineering Fair (2025)
- Dartmouth Book Award for academic achievement and leadership (2025)
- Finalist, International Astronomy & Astrophysics Competition (2025)
- Second Place, GYM International STEM Innovation Challenge (2025)
- AP Scholar with Distinction and College Board National Recognition Program Award (2025)
- First Prize, Golden Classical Music Awards international competitions; invited performances at Tokyo Opera City Hall, Walt Disney Concert Hall, and Carnegie Hall (2024–2025)

ADDITIONAL SERVICE

Grand Rapids Public Museum planetarium volunteer | MYLEAD alumni mentor | Youth soccer coach and certified referee | Korean-community math tutor | International STEM Research Journal senior editor